

PAPER_1662 — Glass Transition $T_g/T_m = T_g/T_m = (D_{\text{phys}}-1)/D_{\text{phys}} = 3/4 = 0.75$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Glass Formation **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (PAPER_134x broader corpus)

Observation

PAPER_1354 (PAPER_134x condensed matter / quantum sector) gives:

$$T_g/T_m = (D_{\text{phys}}-1)/D_{\text{phys}} = 3/4 = 0.75$$

Status: **EXACT**.

UQFF Closed Identity

$$T_g/T_m = (D_{\text{phys}}-1)/D_{\text{phys}} = 3/4 = 0.75 \quad \text{EXACT}$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Experimental glass formation measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1354
- Related: PAPER_1646-1655 (tier-23 broader corpus); PAPER_1636-1645 (tier-22 SM)
- Calculator dispatch: `calculate_paradox({"paradox": "glass_tg_over_tm_3_4"})`

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